Layer 2 Switch CLI Programming Review

Supported commands, runtime state effects, and incompleteness findings

Purpose and conclusion

This review covers the current Layer 2 switch CLI implementation in TSS2. It maps the supported command groups to the records they modify and identifies behavior that is incomplete, misleading, or not yet connected to the simulated network state. The main MVP workflow is present: access VLANs, trunks, allowed and native VLANs, interface shutdown, VLAN creation, show commands, and configuration save.

The most important gaps are default trunk behavior, suspended VLAN presentation, the absence of a real MAC address table, and the fact that switch devices currently accept router-style IP and static-route commands.

Scope and source of truth

- Primary command implementation: Assets/Scripts/TssCliCommandProcessor.cs
- Runtime state records: Assets/Scripts/TssNetworkSessionState.cs
- Change propagation: Assets/Scripts/TssTrainingSession.cs
- Visible terminal path: Assets/Scripts/TssTerminalSessionPresenter.cs
- Existing command coverage: Assets/Tests/EditMode/TssDeviceSessionTests.cs

Command support by mode

EXEC and privileged EXEC

Command group	Support	State effect
enable, en	Supported	Changes NetworkDeviceState.Mode; may set AwaitingEnablePassword.
disable	Supported	Changes mode to UserExec.
configure terminal, conf t	Supported	Changes mode to GlobalConfig.
exit, end	Supported	Changes CLI mode only.
write memory, copy run start	Supported	Clones state into StartupSnapshot and records save evidence.
show running-config	Supported	Reads current device state.
show startup-config	Supported	Reads StartupSnapshot.
show vlan, show vlan brief	Supported	Reads VLAN records and records VLAN evidence.
show interfaces trunk	Supported	Reads trunk records and records trunk evidence.
show interfaces, show ip interface brief	Supported	Reads interface records.
show mac address-table	Partial	Returns a fixed no-dynamic-entries response; no MAC table state exists.
show ip route	Supported but questionable	Reads NetworkRouteState; static routes can be configured on switches.
ping	Placeholder	Always reports 0 percent success; does not evaluate network state.
clear	UI-only	Clears terminal output, not network state.

Global configuration

Command	State record changed
hostname, no hostname	Hostname; also attempts to rename the installed scene or endpoint record.
enable password, no enable password	EnablePassword.
enable secret, no enable secret	EnableSecret.
service password-encryption	PasswordEncryption; affects rendering, not stored-value encryption.
ip domain-lookup, no ip domain-lookup	DomainLookupDisabled.
banner motd, no banner motd	Motd.
line console 0	Selects console and enters LineConfig.
line vty 0 4	Selects VTY and enters LineConfig.
vlan <id>	Creates or selects Vlans[id] and enters VlanConfig.
no vlan <id>	Removes a VLAN database entry, except VLAN 1.
ip route, no ip route	Adds or removes NetworkRouteState records.

interface, interface range	Selects or creates NetworkInterfaceState records.

Access interface commands

Command	State effect
switchport mode access	IsTrunk=false; SwitchportModeConfigured=true.
no switchport mode access	IsTrunk=false; SwitchportModeConfigured=false.
switchport access vlan <id>	Sets AccessVlan and SwitchportAccessVlanConfigured.
no switchport access vlan	Resets AccessVlan to VLAN 1 and clears the configured flag.
shutdown / no shutdown	Sets AdminUp false or true; state change refreshes carriers and endpoint connectivity.
ip address <address> <mask>	Sets Address and Mask, including on switch physical interfaces.
no ip address	Clears Address and Mask.

Trunk interface commands

Command	State effect
switchport trunk encapsulation dot1q	Sets TrunkEncapsulationDot1q=true.
no switchport trunk encapsulation dot1q	Sets TrunkEncapsulationDot1q=false.
switchport mode trunk	IsTrunk=true; SwitchportModeConfigured=true.
no switchport mode trunk	IsTrunk=false; SwitchportModeConfigured=false.
switchport trunk allowed vlan <list>	Replaces AllowedVlans with the supplied VLAN IDs.
no switchport trunk allowed vlan	Clears AllowedVlans.
switchport trunk native vlan <id>	Sets NativeVlan.
no switchport trunk native vlan	Resets NativeVlan to VLAN 1.

Line and VLAN configuration

Mode	Commands	State effect
LineConfig	password, no password	Updates ConsolePassword or VtyPassword.
LineConfig	login, no login	Updates ConsoleLoginRequired or VtyLoginRequired.
VlanConfig	name, no name	Stores the VLAN display name in Vlans[id].
VlanConfig	state suspend, no state suspend	Appends or removes '(suspended)' inside the VLAN name string.

Runtime state behavior

Mutating commands generally pass through Changed(), which calls TssTrainingSession.NotifyNetworkStateChanged(). That refreshes physical carriers, placed desktop connectivity, and the session StateChanged event. Read-only show commands do not call the notification path. Some show commands still mutate evidence sets directly.

Layer 2 availability

For switch interfaces, the runtime evaluator requires the interface to be administratively up, physically up, and attached to an active VLAN. Access ports use AccessVlan. Trunks require at least one active VLAN in AllowedVlans. VLAN suspension is inferred from the text stored in the VLAN name.

Incompleteness findings

Priority	Finding	Why it matters
High	New trunks have no allowed VLANs by default.	AllowedVlans starts empty, and Layer 2 availability requires Any(...). A trunk created with only switchport mode trunk cannot carry traffic until an allowed list is explicitly configured.
High	show vlan always reports active.	The presenter strips '(suspended)' and hardcodes active, so state suspend is not visible in the primary inspection command.
High	Switches accept router-style IP and static-route commands.	ip address and ip route are stored on switch state even though no switch management SVI model exists. This blurs Layer 2 and Layer 3 behavior.

Medium	show interfaces trunk does not filter operational status.	It reports every IsTrunk interface, including administratively down or carrier-down interfaces, despite describing active trunks.
Medium	No dynamic MAC address-table state exists.	show mac address-table cannot display learned MAC addresses, VLANs, or ports.
Medium	ping is not connected to network state.	The command always returns failure and cannot validate switch configuration.
Medium	VLAN suspension is encoded in the VLAN name.	Repeated suspend commands can duplicate the suffix, and status is not represented as a typed field.
Low	Help output is incomplete.	The detailed command reference includes more commands than the mode-specific '?' output, especially no forms and password or banner commands.
Low	CLI mode changes do not notify the training session.	The terminal updates from the returned result, but other state listeners do not receive mode-only changes.

Recommended review order

- Decide whether an empty trunk allowed list means all VLANs or no VLANs in this training simulator.
- Make show vlan display suspended status from the same state used by the connectivity evaluator.
- Decide whether Layer 2 switches should reject ip route and physical-interface ip address, or add an explicit management SVI model.
- Filter show interfaces trunk by AdminUp and CarrierUp when the output is intended to show active trunks.
- Leave MAC learning and functional ping deferred unless they are required for the current training scenario.

Bottom line

The current implementation supports the core switch programming loop: access VLAN assignment, trunk configuration, VLAN creation and naming, interface shutdown, operational inspection, and configuration save. It is sufficient for the existing MVP workflow, but the five high-value review items above should be resolved or explicitly documented before treating the switch CLI as complete.

Review basis

Reviewed October 20 2026 against the current TSS2 source and EditMode command coverage. This document is a source and state-model review; it does not claim a new Play Mode smoke test was run for every command listed.